

ChainGuard 决策报告

摘要

{ "decision_id": "02147cc244dc4a67acfc0f8a26e6e5ac", "timestamp": "2026-07-13T10:09:05.637382+00:00", "event_type": "typhoon_port_shutdown", "event_severity": "高", "inventory_risk_index": 65.99, "constraint_feasible_count": 27, "debate_converged": true, "human_approval_required": false, "decision_status": "ok", "error_message": "", "event_key": "EVT-TYPHOON-NB-001", "net_benefit": 600000.0, "penalty_saving": 180000.0, "profit_protected": 420000.0 }

方案对比

{ "agent_name": "采购 Agent", "role": "供应商替代、采购周期、供货稳定性", "proposal_title": "多源联合补货：B备用供应商", "proposal": "针对 核心控制芯片 的供应缺口，执行“多源联合补货”，计划调动2家合格供应商，优先覆盖关键订单。", "reasoning": ["B备用供应商 当前可供 4200 件，交期成本倍数为 1.45。", "选中策略预计供给 5800 件，自身效用 80.86，系统效用 86.20。", "该策略另含 1 家可供货来源：C第二备用供应商。"], "risks": ["应急采购成本可能高于常规采购。", "供应商可供货量和交期仍需人工确认后才能形成采购承诺。", "单一来源占比过高时仍存在供应连续性风险。"], "actions": ["2小时内确认。", "锁定核心控制芯片的应急采购批次。", "同步联系其他合格供应商，准备分批采购或替代来源。"], "scores": { "timeliness": 75.0, "cost": "****", "risk_reduction": 86.0, "feasibility": 67.06, "service_level": 100.0 }, "strategy_options": [{ "label": "主供应商紧急补货", "own_utility": 75.4, "system_utility": 76.48, "selected": false }, { "label": "多源联合补货", "own_utility": 80.86, "system_utility": 86.2, "selected": true }, { "label": "应急最大化供给", "own_utility": 73.75, "system_utility": 89.96, "selected": false }], "total_score": 77.46, "experience_hints": ["类似案例「台风导致宁波港暂停作业 导致 SUP-A 受影响，库存支撑约」建议：备用供应商 + 关键订单空运 + 非关键订单延期"], "experience_confidence": 0.1 }, { "agent_name": "物流 Agent", "role": "运输时效、路线风险、交付保障", "proposal_title": "最快运输优先：空运", "proposal": "对受影响的 核心控制芯片 订单执行全量空运，在当前可用路线中平衡交付时效和运输成本。", "reasoning": ["台风导致宁波港暂停作业正在影响原有供应链路，需要启用可用替代路线。", "选中策略预计 18 小时到达，自身效用 83.75。", "A类订单最近交付窗口为 72 小时。"], "risks": ["空运的成本倍数约为 3，需核验订单毛利。", "承运能力需要及时确认，否则预计时效可能变化。", "只依赖单一路线中断风险。"], "actions": ["立即锁定空运运力和提货窗口。", "将受影响订单按优先级和交期排序。", "保留其他可用运输方式作为分流和回退方案。"], "scores": { "timeliness": 87.5, "cost": "****", "risk_reduction": 75.0, "feasibility": 75.0, "service_level": 100.0 }, "strategy_options": [{ "label": "最快运输优先", "own_utility": 83.75, "system_utility": 55.0, "selected": true }, { "label": "均衡运输组合", "own_utility": 77.92, "system_utility": 57.75, "selected": false }, { "label": "最低成本路线", "own_utility": 63.33, "system_utility": 69.49, "selected": false }], "total_score": 72.29, "experience_hint": "类似案例「台风导致宁波港暂停作业 导致 SUP-A 受影响，库存支撑约」建议：备用供应商 + 关键订单空运 + 非关键订单延期", "experience_confidence": 0.1 }, { "agent_name": "财务 Agent", "role": "应急成本、订单毛利、违约损失", "proposal_title": "盈亏平衡截止：核心控制芯片", "proposal": "针对 核心控制芯片 执行“盈亏平衡截止”，默认对所有订单使用最高成本方案，而是依据违约损失和订单毛利决定保障范围。", "reasoning": ["A类关键订单 80000，占全部罚金 70.6%。", "选中策略覆盖比例为 100.0%，自身效用 55.44，系统效用 44.39。", "当前订单总毛利为 632000，全部违约罚金为 255000，B/C类订单占比 66.7%。"], "risks": ["若过度压缩应急运输成本，高罚金订单可能出现违约风险。", "若全部使用最高成本方案，部分低毛利订单可能利润转负。", "分客户沟通。"], "actions": ["计算各订单毛利、罚金和应急成本承受上限。", "优先为A类和高罚金订单配置高时效资源。", "对其余订单采用较低成本运输、分批交付。"], "res": { "timeliness": 70.59, "cost": "****", "risk_reduction": 85.44, "feasibility": 80.0, "service_level": 33.33 }, "strategy_options": [{ "label": "A类优先分级保障", "own_utility": 29.35, "system_utility": 28.01, "selected": false }, { "label": "全量高成本保障", "own_utility": 10.44, "system_utility": 21.89, "selected": false }, { "label": "盈亏平衡截止", "own_utility": 55.44, "system_utility": 44.39, "selected": true }], "total_score": 71.67, "experience_hints": ["类似案例「台风导致宁波港暂停作业 导致 SUP-A 受影响，库存支撑约」建议：备用供应商 + 关键订单空运 + 非关键订单延期"], "experience_confidence": 0.1] }

推演要点

{ "conflict": { "has_conflict": true, "conflict_type": "成本-时效冲突", "conflict_summary": "物流主张全量空运，财务反对高成本方案", "highest_agent": "采购 Agent", "lowest_agent": "财务 Agent", "score_gap": 5.79, "reasons": ["方案文本出现冲突关键词：全量空运，利润转负，违约风险。"], "rebuttal": { "debater": "财务 Agent", "target": "物流 Agent", "rebuttal_points": ["最快运输优先：空运成本分值(33/100)偏低，会压缩利润空间。", "在台风导致宁波港暂停作业背景下，高成本运输应聚焦履约损失最大的订单。", "建议仅对1个A类订单合降本。"], "suggested_revision": "将最快运输优先：空运修订为分级保障：优先保障1个A类订单时效，其余订单采用客户沟通与低成本运输组合。", "accepted_trade_off": "接受非关键订单时效让步，换取关键客户履约和整体利润可控。", "llm_used": false, "model_name": "template", "arbitration": { "final_decision_title": "空运 + 备用补货 + 低优先级延期：核心控制芯片应急响应", "final_strategy": "围绕核心控制芯片的断供风险，采用空运 + 备用补货 + 低优先级延期综合策略；触发事件为台风导致宁波港暂停作业。以评分最高方案(多源联合补货：备用供应商)为基准，最终评分为79.5，兼顾交期、成本和风险。", "reasons": ["采纳采购 Agent 的备用供应商建议：优先确认可供货量与交期。", "采纳物流 Agent 的运输方案建议：按关键程度选择运输方式。", "采纳财务 Agent 的成本控制方案：避免高成本方案导致利润转负。"], "rejected_opinions": ["拒绝单一运输方案覆盖全部受影响订单：低毛利订单利润转负风险较高。"], "execution_plan": ["立即联系备用供应商确认核心控制芯片可供货量与交期。", "对1个A类关键客户订单启用优先保障(空运)。", "与非关键客户沟通延期，制定分批交付和替代运输计划。", "若交期紧张，启动第二梯队应急供应预案。"], "manual_confirmation_points": ["是否接受对A类订单的优先保障成本安排(空运)。", "是否启用备用供应商补充核心控制芯片库存。"], "non_key_order_extension_plan": "是否已与客户确认，避免投诉风险。"], "expected_effect": { "supply_continuity": "通过空运 + 备用补货 + 低优先级延期，核心控制芯片供应连续性预计显著改善。", "delivery_risk": "1个A类客户交付风险因资源优先配置而降低。", "cost_risk": "总体成本风险可控，分级保障避免单一高成本方案。", "customer_impact": "1个A类客户优先保障；非关键客户通过提前沟通降低投诉风险。"], "final_score": 79.5 }, "constraint_analysis": { "feasible_count": 27, "feasible": true, "optimal_combo": { "procurement": "应急最大化供给", "logistics": "最低成本路线", "finance": "盈亏平衡截止" }, "optimal_system_utility": 203.84, "individual_system_utility": 185.59, "recommended_changes": ["采购：建议从「多源联合补货」切换到「应急最大化供给」（系统效用+3.76）", "物流：建议从「最快运输优先」切换到「最低成本路线」"], "constraint_violations": [], "all_combos": [{ "cost_multiplier": 4.45, "system_utility": 159.49, "feasible": true, "is_optimal": false, "label": "主供应商紧/最快运输优/A类优先分" }, { "cost_multiplier": 4.45, "system_utility": 153.37, "feasible": true, "is_optimal": false, "label": "主供应商紧/最快运输优/全量高成本" }, { "cost_multiplier": 4.45, "system_utility": 175.87, "feasible": true, "is_optimal": false, "label": "主供应商紧/最快运输优/盈亏平衡截" }, { "cost_multiplier": 3.75, "system_utility": 162.24, "feasible": true, "is_optimal": false, "label": "主供应商紧/均衡运输组/A类优先分" }, { "cost_multiplier": 3.75, "system_utility": 156.12, "feasible": true, "is_optimal": false, "label": "主供应商紧/均衡运输组/全量高成本" }, { "cost_multiplier": 3.75, "system_utility": 178.62, "feasible": true, "is_optimal": false, "label": "主供应商紧/均衡运输组/盈亏平衡截" }, { "cost_multiplier": 2.75, "system_utility": 173.98, "feasible": true, "is_optimal": false, "label": "主供应商紧/最低成本路/A类优先分" }, { "cost_multiplier": 2.75, "system_utility": 159.49, "feasible": true, "is_optimal": false, "label": "主供应商紧/均衡运输组/A类优先分" }] }

ity":167.86,"feasible":true,"is_optimal":false,"label":"主供应商紧/最低成本路/全量高成本"},{"cost_multiplier":2.75,"system_utility":190.36,"feasible":true,"is_optimal":false,"label":"主供应商紧/最低成本路/盈亏平衡截"},{"cost_multiplier":4.381,"system_utility":169.21,"feasible":true,"is_optimal":false,"label":"多源联合补/最快运输优/A类优先分"},{"cost_multiplier":4.381,"system_utility":163.09,"feasible":true,"is_optimal":false,"label":"多源联合补/最快运输优/全量高成本"},{"cost_multiplier":4.381,"system_utility":185.59,"feasible":true,"is_optimal":false,"label":"多源联合补/最快运输优/盈亏平衡截"},{"cost_multiplier":3.681,"system_utility":171.96,"feasible":true,"is_optimal":false,"label":"多源联合补/均衡运输组/A类优先分"},{"cost_multiplier":3.681,"system_utility":165.84,"feasible":true,"is_optimal":false,"label":"多源联合补/均衡运输组/全量高成本"},{"cost_multiplier":3.681,"system_utility":188.34,"feasible":true,"is_optimal":false,"label":"多源联合补/均衡运输组/盈亏平衡截"},{"cost_multiplier":2.681,"system_utility":183.7,"feasible":true,"is_optimal":false,"label":"多源联合补/最低成本路/A类优先分"},{"cost_multiplier":2.681,"system_utility":177.58,"feasible":true,"is_optimal":false,"label":"多源联合补/最低成本路/全量高成本"},{"cost_multiplier":2.681,"system_utility":200.08,"feasible":true,"is_optimal":false,"label":"多源联合补/最低成本路/盈亏平衡截"},{"cost_multiplier":4.251,"system_utility":172.97,"feasible":true,"is_optimal":false,"label":"应急最大化/最快运输优/A类优先分"},{"cost_multiplier":4.251,"system_utility":166.85,"feasible":true,"is_optimal":false,"label":"应急最大化/最快运输优/全量高成本"},{"cost_multiplier":4.251,"system_utility":189.35,"feasible":true,"is_optimal":false,"label":"应急最大化/最快运输优/盈亏平衡截"},{"cost_multiplier":3.551,"system_utility":175.72,"feasible":true,"is_optimal":false,"label":"应急最大化/均衡运输组/A类优先分"},{"cost_multiplier":3.551,"system_utility":169.6,"feasible":true,"is_optimal":false,"label":"应急最大化/均衡运输组/全量高成本"},{"cost_multiplier":3.551,"system_utility":192.1,"feasible":true,"is_optimal":false,"label":"应急最大化/均衡运输组/盈亏平衡截"},{"cost_multiplier":2.551,"system_utility":187.46,"feasible":true,"is_optimal":false,"label":"应急最大化/最低成本路/A类优先分"},{"cost_multiplier":2.551,"system_utility":181.34,"feasible":true,"is_optimal":false,"label":"应急最大化/最低成本路/全量高成本"},{"cost_multiplier":2.551,"system_utility":203.84,"feasible":true,"is_optimal":true,"label":"应急最大化/最低成本路/盈亏平衡截"}]}

执行确认点

["是否接受对A类订单的优先保障成本安排(空运)。", "是否启用备用供应商补充核心控制芯片库存缺口。", "非关键订单延期计划是否已与客户确认，避免投诉"]

审批链

{ "proposalId": "prop-1", "incidentId": "inc-supplier-shutdown", "status": "approved", "riskLevel": "high", "summary": "双供应商加急补货， 预计延误 2 天", "costImpact": 128000.0, "submitter": "供应链负责人", "waitingHours": 1.2, "ccRoleCodes": ["finance"], "transferredTo": null, "countersigned": true, "history": [{ "action": "submit", "userId": "u-scm_lead", "reason": null, "time": "2026-07-13T18:16:25.766431+08:00" }, { "action": "approve", "userId": "u-boss", "reason": null, "time": "2026-07-13T18:19:51.327858+08:00" }, { "action": "countersign", "userId": "u-finance", "reason": null, "time": "2026-07-13T18:23:05.493021+08:00" }], "id": "ap-1", "tenantId": "tenant-demo", "createdAt": "2026-07-13T09:49:22.247325" }, { "proposalId": "prop-1", "incidentId": "inc-supplier-shutdown", "status": "approved", "riskLevel": "high", "summary": "双供应商加急补货， 预计延误 2 天", "costImpact": 128000.0, "submitter": "供应链负责人", "waitingHours": 48.0, "ccRoleCodes": ["finance"], "transferredTo": null, "countersigned": false, "history": [{ "action": "submit", "userId": "u-scm_lead", "time": "2026-07-11T10:25:04.027867+00:00" }, { "action": "approve", "userId": "u-boss", "time": "2026-07-11T10:25:04.027867+00:00" }, { "action": "countersign_timeout_release", "reason": "超时未会签自动放行", "time": "2026-07-13T10:25:04.036893+00:00" }, { "action": "ratify_approve", "userId": "u-finance", "reason": null, "time": "2026-07-13T18:26:36.006540+08:00" }], "id": "ap-timeout-approve", "tenantId": "tenant-demo", "createdAt": "2026-07-13T10:25:04.029175" }, { "proposalId": "prop-1", "incidentId": "inc-supplier-shutdown", "status": "approved", "riskLevel": "high", "summary": "双供应商加急补货， 预计延误 2 天", "costImpact": 128000.0, "submitter": "供应链负责人", "waitingHours": 48.0, "ccRoleCodes": ["finance"], "transferredTo": null, "countersigned": false, "history": [{ "action": "submit", "userId": "u-scm_lead", "time": "2026-07-11T10:25:04.027867+00:00" }, { "action": "approve", "userId": "u-boss", "time": "2026-07-11T10:25:04.027867+00:00" }, { "action": "countersign_timeout_release", "reason": "超时未会签自动放行", "time": "2026-07-13T10:25:04.042909+00:00" }, { "action": "ratify_object", "userId": "u-finance", "reason": "资金依据不足， 需事后复核", "time": "2026-07-13T18:28:35.707276+08:00" }], "id": "ap-timeout-object", "tenantId": "tenant-demo", "createdAt": "2026-07-13T10:25:04.030346" }]